

Problems on H.C.F and L.C.M

1. Find the H.C.F of 108, 288 and 360.

A) 18

B) 36

C) 54

D) 72

Answer: B) 36

2. Find the L.C.M of 24, 36 and 40.

A) 120

B) 240

C) 360

D) 480

Answer: C) 360

3. The H.C.F of two numbers is 11 and their L.C.M is 7700. If one of the numbers is 275, the other is:

A) 279

B) 283

C) 308

D) 318

Answer: C) 308

4. Find the H.C.F of $\frac{2}{3}$, $\frac{8}{9}$, $\frac{64}{81}$ and $\frac{10}{27}$.

A) $\frac{2}{81}$

B) $\frac{2}{3}$

C) $\frac{160}{81}$

D) $\frac{160}{3}$

Answer: A) $\frac{2}{81}$

5. Find the L.C.M of 3, 2.7 and 0.09.

- A) 2.7
- B) 27
- C) 0.27
- D) 2700

Answer: B) 27

6. Two numbers are in the ratio 15:11. If their H.C.F is 13, find the numbers.

- A) 195, 143
- B) 190, 140
- C) 185, 163
- D) 195, 153

Answer: A) 195, 143

7. The L.C.M of two numbers is 48. The numbers are in the ratio 2:3. The sum of the numbers is:

- A) 28
- B) 32
- C) 40
- D) 64

Answer: C) 40

8. Which of the following is a pair of co-primes?

- A) (16, 62)
- B) (18, 25)
- C) (21, 35)
- D) (23, 92)

Answer: B) (18, 25)

9. The product of two numbers is 4107. If the H.C.F of these numbers is 37, then the greater number is:

- A) 101
- B) 107

C) 111

D) 185

Answer: C) 111

10. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

A) 4

B) 7

C) 9

D) 13

Answer: A) 4

11. The H.C.F of two numbers is 23 and the other two factors of their L.C.M are 13 and 14. The larger of the two numbers is:

A) 276

B) 299

C) 322

D) 345

Answer: C) 322

12. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?

A) 4

B) 10

C) 15

D) 16

Answer: D) 16

13. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:

A) 1677

B) 1683

C) 2523

D) 3363

Answer: B) 1683

14. The sum of two numbers is 528 and their H.C.F is 33. The number of pairs of numbers satisfying the above condition is:

A) 4

B) 6

C) 8

D) 12

Answer: A) 4

15. Find the largest number of 4 digits divisible by 12, 15, 18 and 27.

A) 9690

B) 9720

C) 9930

D) 9960

Answer: B) 9720

16. The product of two numbers is 2028 and their H.C.F is 13. The number of such pairs is:

A) 1

B) 2

C) 3

D) 4

Answer: B) 2

17. Find the lowest common multiple of 24, 36 and 40.

A) 120

B) 240

C) 360

D) 480

Answer: C) 360

18. The least number which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4 and 3 is:

- A) 3
- B) 13
- C) 23
- D) 33

Answer: C) 23

19. The H.C.F of two numbers is 8. Which of the following can never be their L.C.M?

- A) 24
- B) 48
- C) 56
- D) 60

Answer: D) 60

20. What is the greatest length which can be used to measure exactly the following lengths: 20 ft, 13 ft 9 inches, 17 ft 6 inches, 21 ft 3 inches?

- A) 9 inches
- B) 1 ft 3 inches
- C) 1 ft 6 inches
- D) 2 ft

Answer: B) 1 ft 3 inches

21. The H.C.F and L.C.M of two numbers are 12 and 336 respectively. If one of the numbers is 84, the other is:

- A) 36
- B) 48
- C) 72
- D) 96

Answer: B) 48

22. Product of two co-prime numbers is 117. Their L.C.M should be:

- A) 1
- B) 117
- C) equal to their H.C.F
- D) cannot be calculated

Answer: B) 117

23. Three numbers are in the ratio 3:4:5 and their L.C.M is 2400. Their H.C.F is:

- A) 40
- B) 80
- C) 120
- D) 200

Answer: A) 40

24. The greatest number which can divide 1356, 1868 and 2764 leaving the same remainder 12 in each case is:

- A) 64
- B) 124
- C) 156
- D) 260

Answer: A) 64

25. The ratio of two numbers is 3:4 and their H.C.F is 4. Their L.C.M is:

- A) 12
- B) 16
- C) 24
- D) 48

Answer: D) 48

26. Find the greatest number that will divide 398, 436 and 542 leaving 7, 11 and 15 as remainders respectively.

A) 11

B) 17

C) 34

D) 45

Answer: B) 17

27. The H.C.F of two numbers is 12 and their difference is 12. The numbers are:

A) 66, 78

B) 70, 82

C) 94, 106

D) 84, 96

Answer: D) 84, 96

28. The LCM of two numbers is 4800 and their HCF is 160. If one of the numbers is 480, then the other number is:

A) 1600

B) 1800

C) 2200

D) 2400

Answer: A) 1600

29. Three different containers contain 496 liters, 403 liters and 713 liters of mixtures of milk and water respectively. What biggest measure can measure all the different quantities exactly?

A) 1 litre

B) 7 litres

C) 31 litres

D) 41 litres

Answer: C) 31 litres

30. If the sum of two numbers is 55 and the H.C.F. and L.C.M. of these numbers are 5 and 120 respectively, then the sum of the reciprocals of the numbers is equal to:

A) 55/601

B) $601/55$

C) $11/120$

D) $120/11$

Answer: C) $11/120$

31. The least multiple of 7, which leaves a remainder of 4, when divided by 6, 9, 15 and 18 is:

A) 74

B) 94

C) 184

D) 364

Answer: D) 364

32. The maximum number of students among them 1001 pens and 910 pencils can be distributed in such a way that each student gets the same number of pens and same number of pencils is:

A) 91

B) 910

C) 1001

D) 1911

Answer: A) 91

33. The H.C.F. of two numbers is 8. Which of the following can never be their L.C.M.?

A) 24

B) 48

C) 56

D) 60

Answer: D) 60

34. L.C.M. of two prime numbers x and y ($x > y$) is 161. The value of $3y - x$ is:

A) -2

B) -1

C) 1

D) 2

Answer: A) -2

35. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:

A) 9000

B) 9400

C) 9600

D) 9800

Answer: C) 9600

36. Find the L.C.M of 72, 108 and 2100.

A) 37800

B) 2100

C) 36600

D) 28400

Answer: A) 37800

37. The H.C.F of 0.54, 1.8 and 7.2 is:

A) 1.8

B) 0.18

C) 0.018

D) 18

Answer: B) 0.18

38. The H.C.F of $\frac{9}{10}$, $\frac{12}{25}$, $\frac{18}{35}$ and $\frac{21}{40}$ is:

A) $\frac{3}{5}$

B) $\frac{252}{5}$

C) $\frac{3}{1400}$

D) $\frac{63}{700}$

Answer: C) $\frac{3}{1400}$

39. Find the least number which when divided by 16, 18, 20 and 25 leaves 4 as remainder in each case, but when divided by 7 leaves no remainder.

A) 17004

B) 18000

C) 18002

D) 18004

Answer: D) 18004

40. Three numbers are in the ratio 2:3:4. The sum of their squares is 1856. The numbers are:

A) 8, 12, 16

B) 16, 24, 32

C) 12, 18, 24

D) None of these

Answer: B) 16, 24, 32

41. The product of two numbers is 1320 and their H.C.F is 6. The L.C.M of the numbers is:

A) 220

B) 1314

C) 1326

D) 7920

Answer: A) 220

42. 252 can be expressed as a product of primes as:

A) $2 \times 2 \times 3 \times 3 \times 7$

B) $2 \times 2 \times 2 \times 3 \times 7$

C) $3 \times 3 \times 3 \times 3 \times 7$

D) $2 \times 3 \times 3 \times 3 \times 7$

Answer: A) $2 \times 2 \times 3 \times 3 \times 7$

43. Find the lowest common multiple of 24, 36 and 40.

A) 120

B) 240

C) 360

D) 480

Answer: C) 360

44. The least number of five digits which is exactly divisible by 12, 15 and 18 is:

A) 10010

B) 10015

C) 10020

D) 10080

Answer: D) 10080

45. The ratio of two numbers is 4:5 and their L.C.M is 120. The numbers are:

A) 30, 40

B) 24, 30

C) 35, 40

D) 40, 50

Answer: B) 24, 30

46. H.C.F of 3240, 3600 and a third number is 36 and their L.C.M is $2^4 \times 3^5 \times 5^2 \times 7^2$. The third number is:

A) $2^2 \times 3^5 \times 7^2$

B) $2^2 \times 5^3 \times 7^2$

C) $2^5 \times 5^2 \times 7^2$

D) $2^3 \times 3^5 \times 7^2$

Answer: A) $2^2 \times 3^5 \times 7^2$

47. Three numbers are in the ratio 1:2:3 and their H.C.F is 12. The numbers are:

A) 6, 12, 18

B) 12, 24, 36

C) 12, 24, 30

D) 5, 10, 15

Answer: B) 12, 24, 36

48. The sum of two numbers is 2000 and their L.C.M is 21879. The two numbers are:

A) 1993, 7

B) 1991, 9

C) 1989, 11

D) 1987, 13

Answer: C) 1989, 11

49. Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.

A) 123

B) 127

C) 235

D) 305

Answer: B) 127

50. Which of the following has the most number of divisors?

A) 99

B) 101

C) 176

D) 182

Answer: C) 176

51. The H.C.F of two numbers is 11 and their L.C.M is 693. If one of the numbers is 77, find the other.

A) 7

B) 9

C) 63

D) 99

Answer: D) 99

52. What is the least number which when divided by 8, 12, 15 and 20 leaves in each case a remainder of 5?

- A) 125
- B) 132
- C) 117
- D) 112

Answer: A) 125

53. The greatest number of 4 digits which is divisible by 15, 25, 40 and 75 is:

- A) 9000
- B) 9400
- C) 9600
- D) 9800

Answer: C) 9600

54. The least number which when divided by 12, 15, 20 and 54 leaves in each case a remainder of 8 is:

- A) 504
- B) 536
- C) 544
- D) 548

Answer: D) 548

55. H.C.F of $4 \times 27 \times 3125$, $8 \times 9 \times 25 \times 7$ and $16 \times 81 \times 5 \times 11 \times 49$ is:

- A) 180
- B) 360
- C) 540
- D) 1260

Answer: A) 180

56. Find the side of the largest possible square slabs which can be paved on the floor of a room 2m 50cm long and 1m 50cm broad.

- A) 40 cm
- B) 50 cm
- C) 55 cm
- D) 60 cm

Answer: B) 50 cm

57. The H.C.F of two numbers is 12 and their difference is 12. Which of the following can be the numbers?

- A) 66, 78
- B) 70, 82
- C) 94, 106
- D) 84, 96

Answer: D) 84, 96

58. The product of two numbers is 1280 and their H.C.F is 8. The L.C.M of the numbers will be:

- A) 128
- B) 160
- C) 150
- D) 140

Answer: B) 160

59. The least perfect square number divisible by 3, 4, 5, 6 and 8 is:

- A) 900
- B) 1200
- C) 2500
- D) 3600

Answer: D) 3600

60. The sum of two numbers is 45. Their difference is $\frac{1}{9}$ of their sum. Their L.C.M is:

- A) 100

B) 150

C) 200

D) 250

Answer: A) 100

61. What is the smallest number which leaves a remainder of 3 when divided by any of the numbers 5, 6 or 8 but leaves no remainder when it is divided by 9?

A) 123

B) 603

C) 723

D) 243

Answer: D) 243

62. Two numbers are in the ratio 3:5. Their L.C.M is 225. The smaller number is:

A) 45

B) 60

C) 75

D) 90

Answer: A) 45

63. The sum of two numbers is 36 and their H.C.F is 4. How many pairs of such numbers are possible?

A) 1

B) 2

C) 3

D) 4

Answer: C) 3

64. The least number which when divided by 48, 60, 72, 108 and 140 leaves 38, 50, 62, 98 and 130 as remainders respectively is:

A) 11115

B) 15110

C) 15120

D) 15210

Answer: B) 15110

65. Find the greatest number that will divide 964, 1238 and 1400 leaving remainders of 41, 31 and 51 respectively.

A) 61

B) 71

C) 73

D) 81

Answer: B) 71

66. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds, all starting at the same point. After what time will they meet again at the starting point?

A) 26 minutes 18 seconds

B) 42 minutes 36 seconds

C) 45 minutes

D) 46 minutes 12 seconds

Answer: D) 46 minutes 12 seconds

67. The H.C.F and L.C.M of two numbers are 50 and 250 respectively. If the first number is divided by 2, the quotient is 50. The second number is:

A) 50

B) 100

C) 125

D) 250

Answer: C) 125

68. The product of two numbers is 1320 and their H.C.F is 6. The L.C.M of the numbers is:

A) 220

B) 1314

C) 1326

D) 7920

Answer: A) 220

69. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:

A) 1677

B) 1683

C) 2523

D) 3363

Answer: B) 1683

70. Find the largest number which divides 62, 132 and 237 to leave the same remainder in each case.

A) 8

B) 35

C) 57

D) 75

Answer: B) 35

71. Find the L.C.M of 3, 2.7 and 0.09.

A) 2.7

B) 27

C) 0.27

D) 2700

Answer: B) 27

72. What is the smallest number which when increased by 5 is completely divisible by 8, 11 and 24?

A) 264

B) 259

C) 269

D) 250

Answer: B) 259

73. The H.C.F of two numbers is 8. Which of the following can never be their L.C.M?

A) 24

B) 48

C) 56

D) 60

Answer: D) 60

74. If the L.C.M of three numbers is 9570, then their H.C.F will be:

A) 11

B) 12

C) 19

D) 21

Answer: A) 11

75. Three numbers are in the ratio 2:3:4 and their H.C.F is 12. The L.C.M of the numbers is:

A) 144

B) 192

C) 240

D) 72

Answer: A) 144

76. The sum of two numbers is 462 and their highest common factor is 22. What is the minimum number of pairs that satisfy these conditions?

A) 2

B) 3

C) 4

D) 6

Answer: D) 6

77. Find the greatest number of five digits which is divisible by 15, 21 and 36.

A) 99640

B) 99540

C) 99720

D) 99270

Answer: B) 99540

78. 105 goats, 140 donkeys and 175 cows have to be taken across a river. There is only one boat which will have to make many trips. The lazy boatman has his own conditions. He insists that he will take the same number of animals in every trip and they have to be of the same kind. He will naturally like to take the largest possible number each time. Can you tell how many animals went in each trip?

A) 35

B) 70

C) 25

D) 15

Answer: A) 35

79. Find the L.C.M of 148 and 185.

A) 680

B) 740

C) 2960

D) 3700

Answer: B) 740

80. The sum of two numbers is 84 and their H.C.F is 12. Total number of such pairs of numbers is:

A) 2

B) 3

C) 4

D) 5

Answer: B) 3

81. The LCM of two numbers is 20 times their HCF. The sum of HCF and LCM is 2520. If one of the numbers is 480, then the other number is:

- A) 400
- B) 480
- C) 520
- D) 600

Answer: D) 600

82. Three planets revolve round the Sun once in 200, 250 and 300 days respectively in their own orbits. When do they all come relatively to the same position as at a certain point of time in their orbits?

- A) After 3000 days
- B) After 2000 days
- C) After 1500 days
- D) After 1200 days

Answer: A) 3000 days

83. The greatest number which can divide 284, 698 and 1618 leaving the same remainder 8 in each case is:

- A) 46
- B) 23
- C) 92
- D) 138

Answer: A) 46

84. H.C.F of two numbers is 13. If these two numbers are in the ratio of 15:11, then find the numbers.

- A) 230, 140
- B) 215, 130
- C) 195, 143
- D) 155, 115

Answer: C) 195, 143

85. Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11 respectively.

A) 12

B) 10

C) 16

D) 18

Answer: A) 12

86. The LCM of two numbers is 495 and their HCF is 5. If the sum of the numbers is 100, then their difference is:

A) 10

B) 46

C) 70

D) 90

Answer: A) 10

87. Two numbers are in the ratio 3:5 and their L.C.M is 75. The H.C.F is:

A) 3

B) 5

C) 10

D) 15

Answer: B) 5

88. Find the side of the largest possible square slabs which can be paved on the floor of a room 5m 44cm long and 3m 74cm broad.

A) 34 cm

B) 44 cm

C) 56 cm

D) 64 cm

Answer: A) 34 cm

89. A rectangular courtyard 3.78 meters long and 5.25 meters wide is to be paved exactly with square tiles, all of the same size. What is the largest size of the tile which could be used for the purpose?

- A) 14 cm
- B) 21 cm
- C) 42 cm
- D) 28 cm

Answer: B) 21 cm

90. The product of two numbers is 4107. If the H.C.F of these numbers is 37, then the greater number is:

- A) 101
- B) 107
- C) 111
- D) 185

Answer: C) 111

91. The sum of two numbers is 36 and their H.C.F is 4. How many pairs of such numbers are possible?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C) 3

92. The least number which when increased by 5 is divisible by each one of 24, 32, 36 and 54 is:

- A) 427
- B) 859
- C) 869
- D) 4320

Answer: B) 859

93. Find the largest number which divides 25, 73 and 97 to leave the same remainder in each case.

A) 24

B) 21

C) 23

D) 6

Answer: A) 24

94. The greatest possible length which can be used to measure exactly the lengths 7 m, 3 m 85 cm, 12 m 95 cm is:

A) 15 cm

B) 25 cm

C) 35 cm

D) 42 cm

Answer: C) 35 cm

95. Three numbers are in the ratio of 3:4:5 and their L.C.M. is 2400. Their H.C.F. is:

A) 40

B) 80

C) 120

D) 200

Answer: A) 40

96. If the sum of two numbers is 36 and their H.C.F. and L.C.M. are 3 and 105 respectively, the sum of the reciprocals of the two numbers is:

A) $\frac{2}{35}$

B) $\frac{3}{35}$

C) $\frac{4}{35}$

D) $\frac{2}{25}$

Answer: C) $\frac{4}{35}$

97. Find the H.C.F of 513, 1134 and 1215.

A) 18

B) 27

C) 33

D) 36

Answer: B) 27

98. The L.C.M of two numbers is 45 times their H.C.F. If one of the numbers is 125 and the sum of H.C.F and L.C.M is 1150, the other number is:

A) 215

B) 220

C) 225

D) 235

Answer: C) 225

99. The greatest number that exactly divides 105, 1001 and 2436 is:

A) 3

B) 7

C) 11

D) 21

Answer: B) 7

100. Find the least number which is a multiple of 31 and when divided by 15, 24 and 32 leaves the remainders 2, 11 and 19 respectively.

A) 2356

B) 2387

C) 2325

D) 2418

Answer: B) 2387